SPT(Part-2)

Spanning Tree Protocol

- Topics covered:
  - STP States/Timers
  - STP BPDU
  - STP Optional Features
  - STP Configuration

Spanning Tree Port States

◆ Stable and Transitional States

STP Port State	Stable/Transitional
Blocking	Stable
Listening	Transitional
Learning	Transitional
Forwarding	Stable

- BLOCKING and FORWARDING are stable states
- LISTENING and LEARNING are transitional states

◆ Stable States Explanation

- Root ports and Designated ports stay stable in Forwarding
- Non-designated ports stay stable in Blocking
- Ports stay stable only if the topology does not change
- If:
  - A new device is added
  - An interface is shut down

- A hardware failure occurs
  → ports may change states

◆ Transitional States Explanation

- Listening and Learning are passed through:
  - When an interface is activated
  - When a Blocking port must move to Forwarding

◆ Disabled State

- Refers to an interface that is **administratively disabled (shutdown)**
- Does not play a role in Spanning Tree

◆ Blocking State

STP Port State	Stable/Transitional
Blocking	Stable

- Non-designated ports are in Blocking state
- Purpose:
  - Prevent loops
  - Disable redundant paths

Behaviour in Blocking state:

- Does NOT send or receive regular traffic
- Drops all regular traffic
- Receives STP BPDUs
- Does NOT forward BPDUs
- Does NOT learn MAC addresses

[Blocking Port]

- Regular Traffic: DROPPED

- BPDU Receive: YES

- BPDU Forward: NO

- MAC Learning: NO

◆ Listening State

STP Port State	Stable/Transitional
Listening	Transitional

- Only Root ports and Designated ports enter Listening
- Non-designated ports never enter Listening
- Duration: 15 seconds (Forward Delay timer)

Behaviour in Listening state:

- Sends and receives only STP BPDUs
- Does NOT forward regular traffic
- Does NOT learn MAC addresses
- Regular traffic is discarded

[Listening Port]

- BPDU Send/Receive: YES

- Regular Traffic: NO

- MAC Learning: NO

◆ Learning State

STP Port State	Stable/Transitional
Learning	Transitional

- Comes after Listening
- Only Root ports and Designated ports
- Duration: 15 seconds (Forward Delay timer)

Behaviour in Learning state:

- Sends and receives only STP BPDUs
- Does NOT forward regular traffic
- Learns MAC addresses

[Learning Port]

- BPDU Send/Receive: YES

- Regular Traffic: NO

- MAC Learning: YES

  - Total time from Listening → Learning → Forwarding:
    - 15 + 15 = 30 seconds

◆ **Forwarding State**

STP Port State	Stable/Transitional
Forwarding	Stable

- Root and Designated ports are stable here
- Port operates normally

Behaviour in Forwarding state:

- Sends and receives BPDUs
- Sends and receives regular traffic
- Learns MAC addresses

[Forwarding Port]

- BPDU Send/Receive: YES

- Regular Traffic: YES

- MAC Learning: YES

- **Spanning Tree Port State Summary**

STP Port State	Send/Receive BPDUs	Frame forwarding (regular traffic)	MAC address learning	Stable/ Transitional
Blocking	NO/YES	NO	NO	Stable
Listening	YES/YES	NO	NO	Transitional
Learning	YES/YES	NO	YES	Transitional
Forwarding	YES/YES	YES	YES	Stable
Disabled	NO/NO	NO	NO	Stable

- **Spanning Tree Timers**

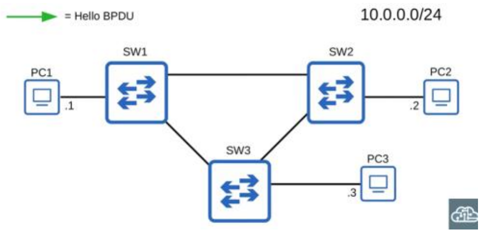
STP Timer	Purpose	Duration
Hello	How often the root bridge sends hello BPDUs	2sec
Forward delay	How long the switch will stay in the Listening and Learning states (each state is 15 seconds = total 30 seconds)	15sec
Max Age	How long an interface will wait after ceasing to receive Hello BDPUs to change the STP topology.	20sec (10* hello)

- **Hello Timer**
  - Determines how often root bridge sends hello BPDUs
  - Default: every 2 seconds
  - Other switches:
    - Do not originate BPDUs
    - Only forward received BPDUs

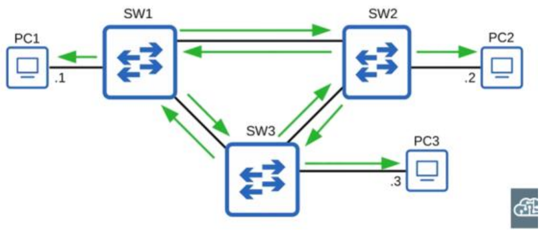
BPDU Forwarding Rule:

- Switches only forward BPDUs on Designated ports
- They do NOT forward BPDUs on:
  - Root ports
  - Non-designated ports

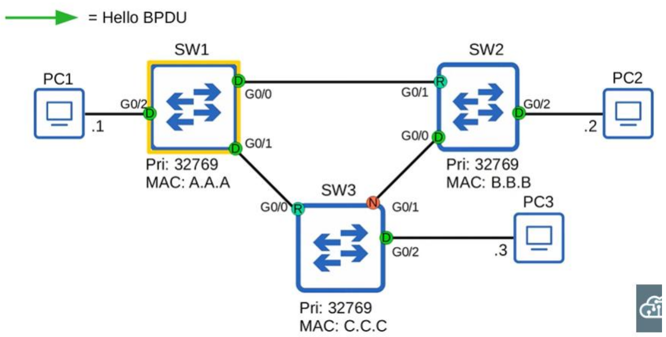
BPDU Flow Example (Initial State)



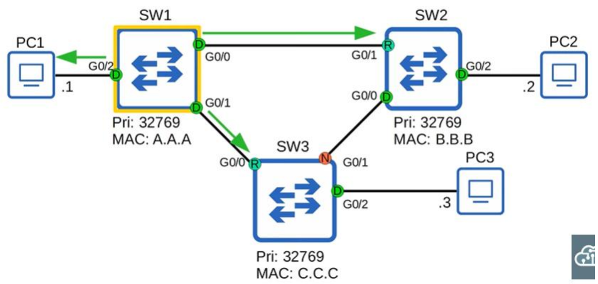
1. All switches think they are root
2. All send BPDUs on all ports



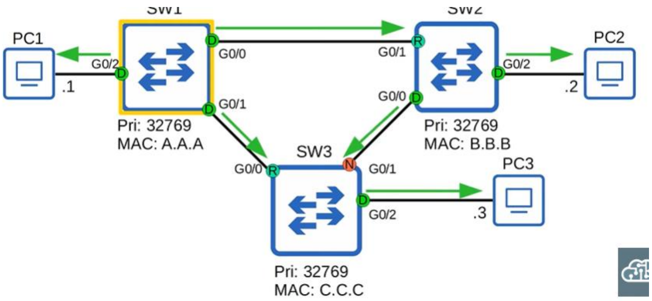
BPDU Flow After Convergence



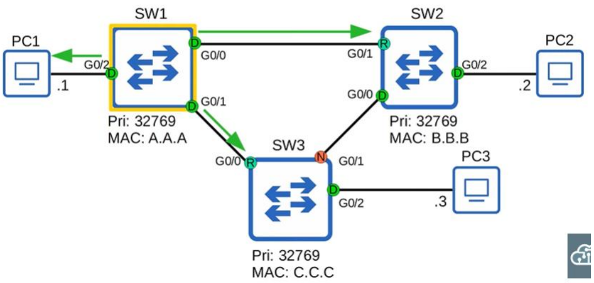
1. Root Bridge sends BPDU every 2 sec
2. Root→[Designated Ports only forward BPDU]



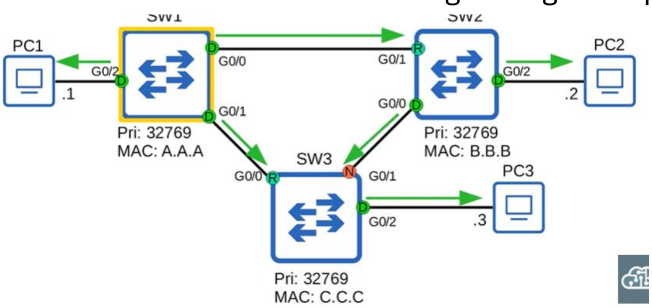
3. Other switches only forward on Designated port only



4. Information updated:
    - Root cost
    - Sending bridge ID
    - Sending port ID
5. Then every 2 seconds later
    - Root bridge will send BPDUs again



    - Other switches will forward through designated port



6. Note switches do not forward the BPDUs out of their root ports and non-designated port, only designated ports.

SO THAT'S THE HELLO TIMER

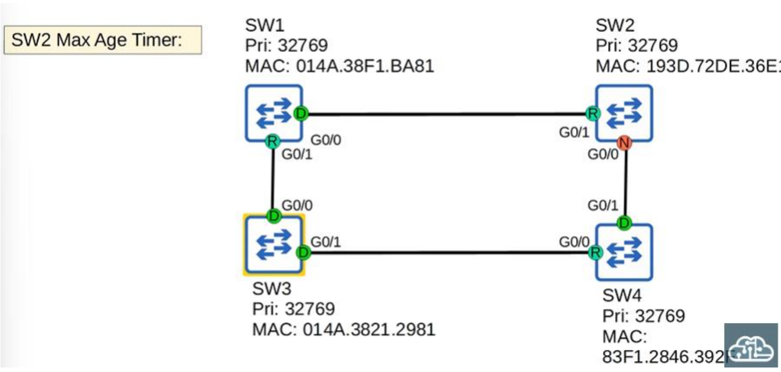
Forward Delay Timer

- The forward delay timer is the length of the Listening and Learning transitional states.
- This timer applies when a port moves toward the forwarding state.
- This is the length of each state, not the total combined time.
- Listening = 15 seconds
- Learning = 15 seconds
- Total time = 30 seconds
- With the default forward delay = 15 seconds:
  - A switchport takes 30 seconds total to move through Listening and Learning and begin forwarding traffic.

Max Age Timer

- The max age timer indicates how long an interface will wait to change the spanning tree topology after it stops receiving BPDUs.

Network Topology:



- SW3 is the root bridge
- SW3 sends BPDUs
- SW1 and SW4 forward BPDUs out of their designated ports
- Each collision domain has one designated port.
- BPDUs are forwarded only out of designated ports.

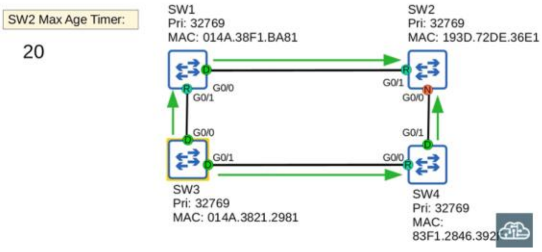
Therefore:

- Root ports expect to receive BPDUs
- Non-designated ports expect to receive BPDUs

Max Age Timer Countdown (Normal Operation)

Focus: SW2 – G0/1 Interface:

- SW2 receives a BPDU



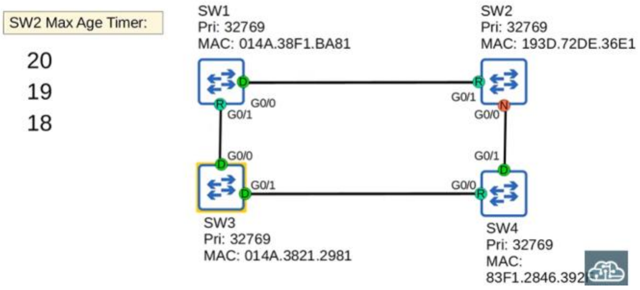
- Max Age timer is reset to 20

  20

  19

  18

- After 2 seconds (hello timer):



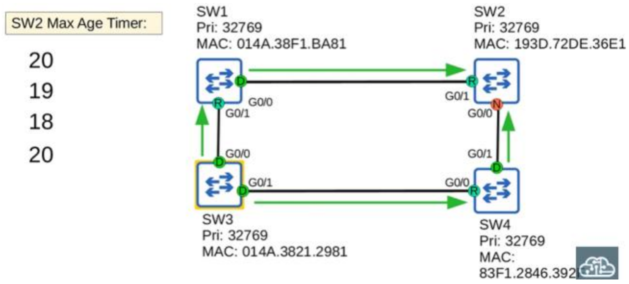
  - Root bridge sends BPDUs again
  - Other switches forward them
  - SW2 receives BPDU again
  - Max Age resets to **20**

    20

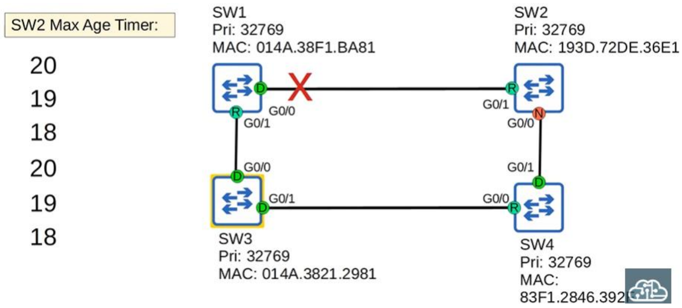
    19

    18

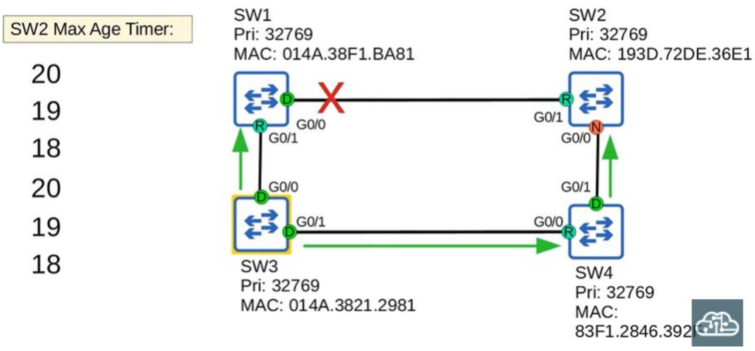
20



Link Failure Scenario



- Failure occurs between SW1 and SW2
- Root bridge still sends BPDUs
- Other switches forward BPDUs
- SW2 no longer receives BPDU on G0/1



Max Age Countdown (Failure)

- Max Age timer continues counting down

  17

  16

  15

  ...

0

Max Age Timer Result

Case 1: BPDU Received Before Timer Reaches 0

- Max Age timer resets to 20
- No topology change occurs

Case 2: No BPDU Received (Timer Reaches 0)

- Switch reevaluates spanning tree
- Decisions include:
  - Root bridge
  - Root ports
  - Designated ports
  - Non-designated ports

Blocking to Forwarding Transition

- If a non-designated port is selected to become:
  - Root port or
  - Designated port

  Then it transitions:

  Blocking

  ↓

  Listening (15 sec)

  ↓

  Learning (15 sec)

  ↓

  Forwarding

  - Total time:
    - 20 sec (Max Age)

    - 15 sec (Listening)
    - 15 sec (Learning)

  Total = 50 seconds

Reason for Long Timers

- These timers exist to make sure loops are not accidentally created
- Layer 2 loops are dangerous
- Spanning Tree is careful before moving a port to forwarding
- A forwarding port can move directly to blocking
- A blocking port cannot move directly to forwarding

Spanning Tree BPDU

Ethernet Header

```
> Frame 999: 68 bytes on wire (544 bits), 68 bytes captured (544 bits) on interface 0
> Ethernet II, Src: aa:aa:aa:aa:aa:ab (aa:aa:aa:aa:aa:ab), Dst: PVST+ (01:00:0c:cc:cc:cd)
> 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 10
> Logical-Link Control
v Spanning Tree Protocol
    Protocol Identifier: Spanning Tree Protocol (0x0000)
    Protocol Version Identifier: Spanning Tree (0)
    BPDU Type: Configuration (0x00)
  v BPDU flags: 0x00
      0... .... = Topology Change Acknowledgment: No
      .... ...0 = Topology Change: No
  v Root Identifier: 32768 / 10 / aa:aa:aa:aa:aa:aa
      Root Bridge Priority: 32768
      Root Bridge System ID Extension: 10
      Root Bridge System ID: aa:aa:aa:aa:aa:aa (aa:aa:aa:aa:aa:aa)
    Root Path Cost: 0
  v Bridge Identifier: 32768 / 10 / aa:aa:aa:aa:aa:aa
      Bridge Priority: 32768
      Bridge System ID Extension: 10
      Bridge System ID: aa:aa:aa:aa:aa:aa (aa:aa:aa:aa:aa:aa)
    Port identifier: 0x8002
    Message Age: 0
    Max Age: 20
    Hello Time: 2
    Forward Delay: 15
```

Destination MAC Address

```
> Frame 999: 68 bytes on wire (544 bits), 68 bytes captured (544 bits) on interface 0
> Ethernet II, Src: aa:aa:aa:aa:aa:ab (aa:aa:aa:aa:aa:ab), Dst: PVST+ (01:00:0c:cc:cc:cd)
```

- Cisco PVST+ BPDU destination MAC:

  0100.0ccc.cccd

- Regular (non-Cisco) STP destination MAC:

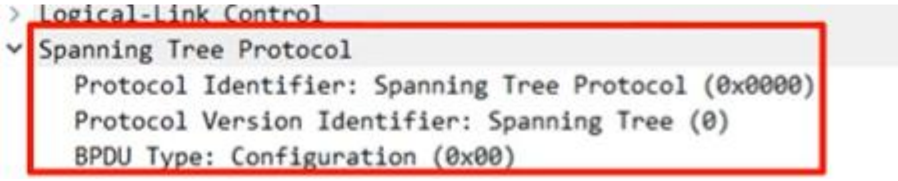
  0180.c200.0000

PVST vs PVST+

- PVST
  - Older
  - Uses ISL
- PVST+
  - Newer
  - Supports dot1q

- ISL is almost never used anymore

BPDU Fields (Overview)



Protocol Identifier = 0000

Protocol Version = 0

BPDU Type = 00 (Configuration BPDU)

Flags

- Used to signal topology changes
- Not required in depth for CCNA

Root Identifier

Includes:

- Bridge Priority
- Extended System ID (VLAN ID)
- Bridge System ID (MAC address)

Root Path Cost = 0 → Root bridge

Bridge Identifier

- Same information as Root Identifier
- Confirms this switch is the root bridge

Port Identifier

0x8002

- 80 (hex) = 128 → default port priority
- 02 → port number

Message Age

- Starts at **0** at the root bridge
- Increases by **1** at each hop
- Subtracted from max age when BPDU is received

Timers in BPDU

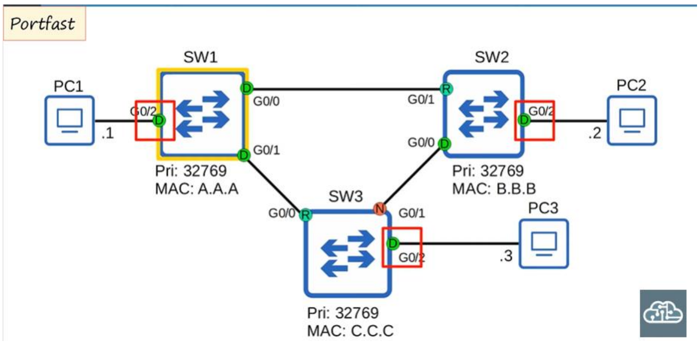
- Max Age
- Hello Time
- Forward Delay
- Timers on the **root bridge** determine timers for all switches

Spanning Tree Toolkit

PortFast

- PortFast solves one problem of spanning tree
- Enabled on interfaces connected to end hosts

  PC ---- Switch (G0/2)



- These ports are designated ports in forwarding
- But normally must wait:
  - Listening (15 sec)
  - Learning (15 sec)

PortFast Experiment (Packet Tracer)

- Enable Show Link Lights
- Connect PC to switch
- Observe link light:

  Orange → Listening/Learning

  Green → Forwarding

- Takes 30 seconds without PortFast

Why This Delay Exists

- Layer 2 loops are dangerous
- Switch must be sure no loop exists
- Only links between switches can form loops
- End hosts cannot form loops

What PortFast Does

- PortFast allows a port to:
    - Move immediately to Forwarding
    - Bypass Listening and Learning

      Access Port + End Host

      → Immediate Forwarding

PortFast Rules

- Enable **only on ports connected to end hosts**
- Do NOT enable on ports connected to switches
- Otherwise, a Layer 2 loop can occur

PortFast Configuration (Interface Mode)

```
SW1(config)#interface g0/2
SW1(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
 host. Connecting hubs, concentrators, switches, bridges, etc... to this
 interface  when portfast is enabled, can cause temporary bridging loops.
 Use with CAUTION

%Portfast has been configured on GigabitEthernet0/2 but will only
 have effect when the interface is in a non-trunking mode.
SW1(config-if)#
```

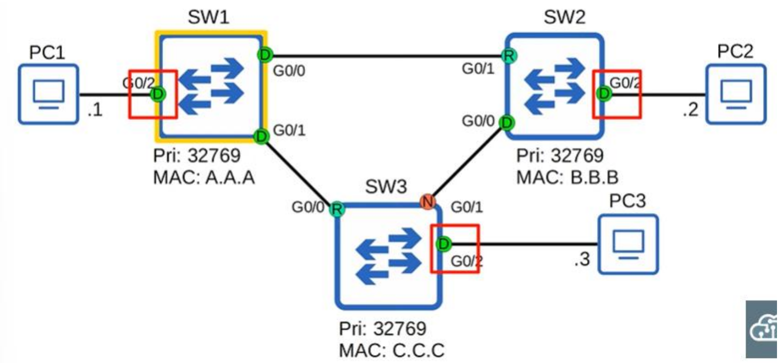
- Warning displayed: Use only on ports connected to a single host
- PortFast works only on non-trunk (access) ports

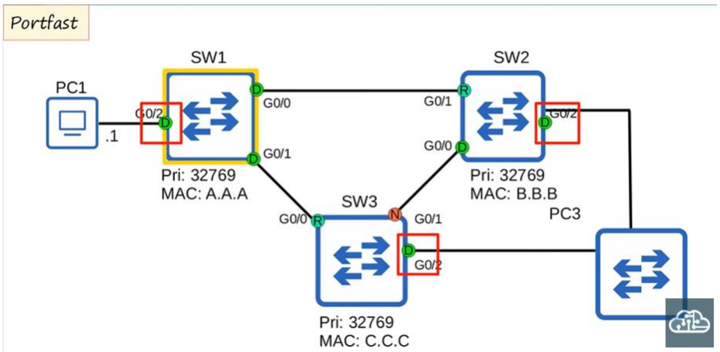
Global PortFast Configuration

"spanning-tree portfast default"

- Enables PortFast on all access ports
- Does not enable on trunk ports

PortFast Risk Example





- Loop is formed immediately
- Can also happen due to:
    - Cabling changes
    - Accidental switch connection

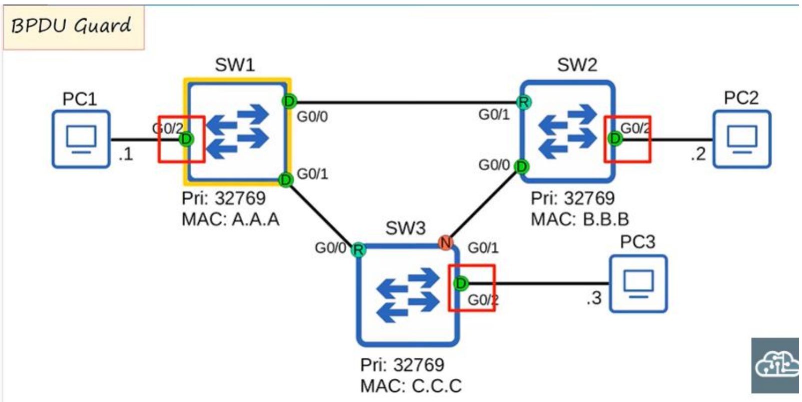
Summary

- PortFast speeds up end-host connectivity
- Removes 30-second delay
- Comes with risk if used incorrectly
- Must be used carefully

BPDU Guard

- There is an additional optional spanning tree feature to protect against loops.
- It is called BPDU Guard.
- If an interface with BPDU Guard enabled receives a BPDU from another switch, the interface will be shut down to prevent a loop.

BPDU Guard – Topology Example



- BPDU Guard is enabled on interfaces connected to **end hosts**
- If a switch is connected instead of a PC, a BPDU is received

BPDU Guard Configuration (Interface Level)

```
SW1(config)#interface g0/2
SW1(config-if)#spanning-tree bpduguard enable
SW1(config-if)#
```

- BPDU Guard is enabled directly on the interface
- If a BPDU is received, the port is disabled

BPDU Guard – Global Configuration

"spanning-tree portfast bpduguard default"

- This enables BPDU Guard on all portfast-enabled interfaces
- Command difference:
    - Interface level: spanning-tree bpduguard enable

- Global level: spanning-tree portfast bpduguard default

BPDU Guard in Action (Packet Tracer)

I took this screenshot in packet tracer, But I connected a switch to a BPDU-Guard enable interface and now you can see what happens when a BPDU arrives on a BPUD-Guard enable port, The port is disable it is effectively shutdown

```
%SPANTREE-2-BLOCK_BPDUGUARD: Received BPDU on port FastEthernet0/1 with BPDU Guard
enabled. Disabling port.

%PM-4-ERR_DISABLE: bpduguard error detected on 0/1, putting 0/1 in err-disable state

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
```

- A BPDU arrives on a BPDU Guard-enabled port
- The port is disabled (err-disabled)

Re-enabling a BPDU Guard Disabled Port

```
Switch(config-if)#shutdown

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to administratively down
Switch(config-if)#no shutdown

Switch(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%SPANTREE-2-BLOCK_BPDUGUARD: Received BPDU on port FastEthernet0/1 with BPDU Guard enabled. Disabling port.

%PM-4-ERR_DISABLE: bpduguard error detected on 0/1, putting 0/1 in err-disable state

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
```

- The interface comes back up
- If still connected to a switch:
  - It will immediately be disabled again
- The problem must be fixed first

Optional Spanning Tree Features (Overview)

- Only PortFast is clearly listed in exam topics
- BPDU Guard is closely related to PortFast
- Other features exist, but are not required in depth

Root Guard

- If Root Guard is enabled on an interface:
    - Even if it receives a superior BPDU
    - The switch will not accept a new root bridge
- The interface is disabled
- Helps protect the intended root bridge

Loop Guard

- If Loop Guard is enabled on an interface:
    - Even if the interface stops receiving BPDUs
    - It will not start forwarding
- The interface is disabled
- Prevents loops caused by unidirectional links

Spanning Tree Mode Configuration

Configure the Spanning Tree mode

```
SW1(config)#spanning-tree mode ?
  mst          Multiple spanning tree mode
  pvst         Per-Vlan spanning tree mode
  rapid-pvst   Per-Vlan rapid spanning tree mode

SW1(config)#spanning-tree mode pvst
```

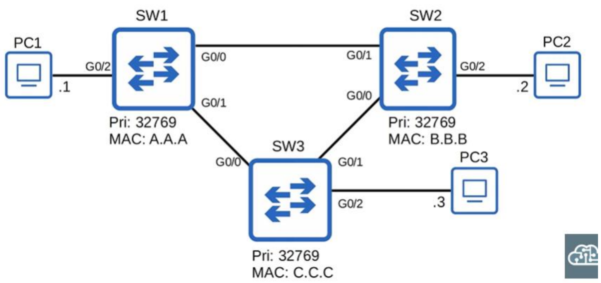
Available options:

- **mst** – Multiple Spanning Tree (not required for CCNA)
- **pvst** – Per-VLAN Spanning Tree
- **rapid-pvst** – Improved version (default on modern switches)

Enabling Classic PVST Mode:

- "spanning-tree mode pvst"
- Used for lab demonstrations
- Rapid-PVST is default and usually not changed

Root Bridge Selection Example



- With default priorities, **SW1** is the root bridge
- We can manually change the root bridge
- Also one primary and secondary root bridge

- **Configure Primary Root Bridge**

```
SW3(config)#spanning-tree vlan 1 root primary
SW3(config)#do show spanning-tree
```

- “spanning-tree vlan vlan-number root primary”
- This command Sets STP priority to **24576**
- If another switch already has a lower priority 24576:
    - It Sets switch priority to **4096 less** than other switch priority
- Ensures this switch becomes root

Resulting Configuration:

```
SW3(config)#spanning-tree vlan 1 root primary
SW3(config)#do show spanning-tree

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    24577
             Address     cccc.cccc.cccc
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24577  (priority 24576 sys-id-ext 1)
             Address     cccc.cccc.cccc
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  15  sec
```

“spanning-tree vlan 1 priority 24576”

- This is the actual command applied
- Root primary is a shortcut for priority configuration

If you then check running config:

```
!
spanning-tree mode pvst
spanning-tree extend system-id
spanning-tree vlan 1 priority 24576
!
```

- **Configure Secondary Root Bridge**

```
SW2(config)#spanning-tree vlan 1 root secondary
SW2(config)#do show spanning-tree

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    24577
             Address     cccc.cccc.cccc
             Cost        4
             Port        1 (GigabitEthernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    28673  (priority 28672 sys-id-ext 1)
             Address     bbbb.bbbb.bbbb
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec
```

"spanning-tree vlan 1 root secondary"

- Sets priority to **28672**
- This switch becomes second-lowest priority
- Will become root if primary fails

Secondary Root Priority Result:

"spanning-tree vlan 1 priority 28672"

- Like root primary, this is also a shortcut
- Actual configuration uses priority values

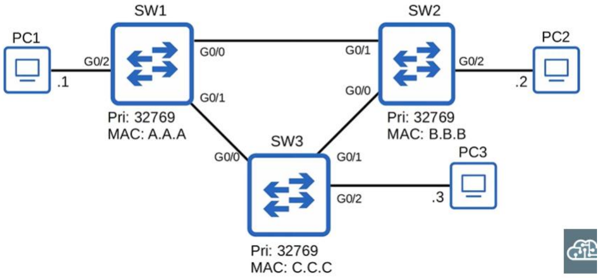
If you check running config:

```
!
spanning-tree mode pvst
spanning-tree extend system-id
spanning-tree vlan 1 priority 28672
!
```

- **Root Bridge Priority Rule**
  - Bridge priority must be set in increments of 4096
  - Root commands are easier than remembering values
  - Root primary and root secondary simplify configuration

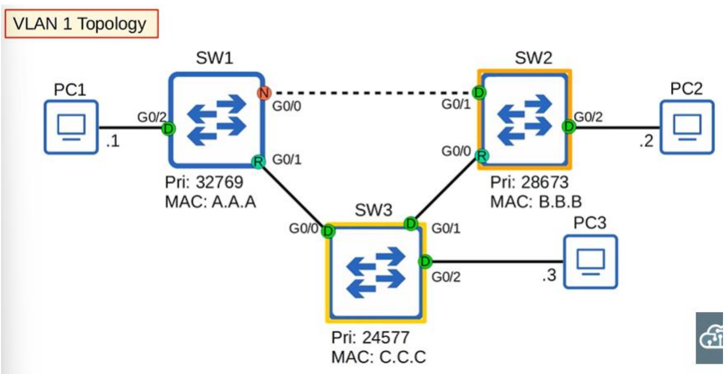
STP Load-Balancing

- **Old Topology:**



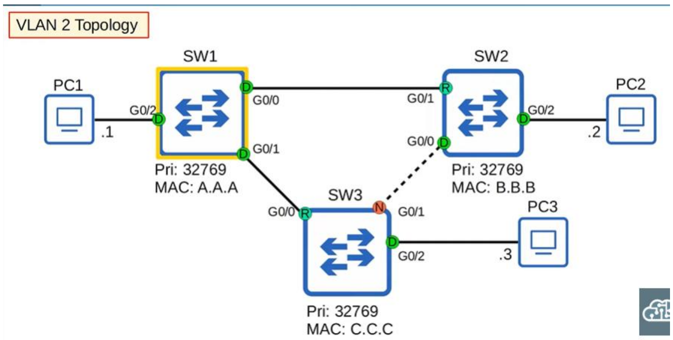
Current Topology:

- This is the topology now.
- The interface between SW1 and SW2 is disabled.
- This is because SW1 is blocking its G0/0 interface.
- The network is running Cisco PVST+.
- This topology applies only to VLAN 1.



VLAN 2 Topology

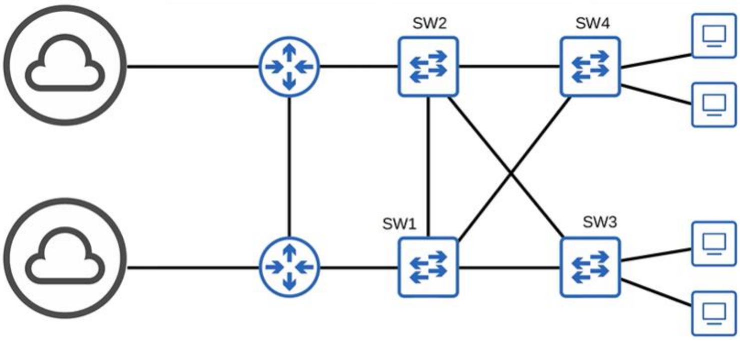
- Assume another VLAN exists: VLAN 2
- The root bridge settings were applied only to VLAN 1
- VLAN 2 uses the default topology
- The link between SW1 and SW2 is NOT blocked
- Instead, the link between SW2 and SW3 is blocked



Spanning Tree Load Balancing

- Blocking the same interface in every VLAN is a waste
- That interface does nothing unless a failure occurs
- By using different root bridges per VLAN:
    - Different VLANs block different links
- This spreads traffic across interfaces
- This is called spanning tree load balancing

◆ **Quiz Question 7 (Concept)**



- Two VLANs are active: **VLAN 10** and **VLAN 20**
- By default, **SW3 is the root bridge for both**
- Requirements:
    - SW1 → Primary root for VLAN 10, Secondary for VLAN 20
    - SW2 → Primary root for VLAN 20, Secondary for VLAN 10

Commands for SW1

spanning-tree vlan 10 root primary

spanning-tree vlan 20 root secondary

- SW1 is the root bridge for VLAN 10
- SW1 is backup root for VLAN 20

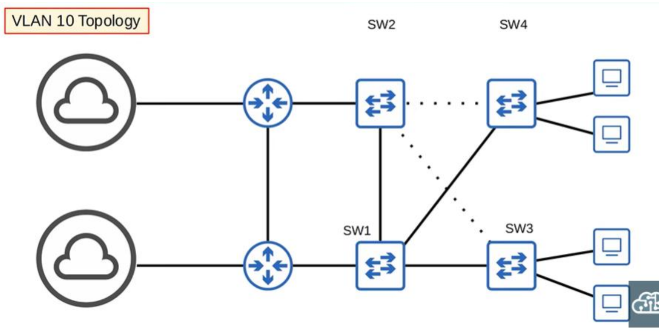
Commands for SW2

spanning-tree vlan 20 root primary

spanning-tree vlan 10 root secondary

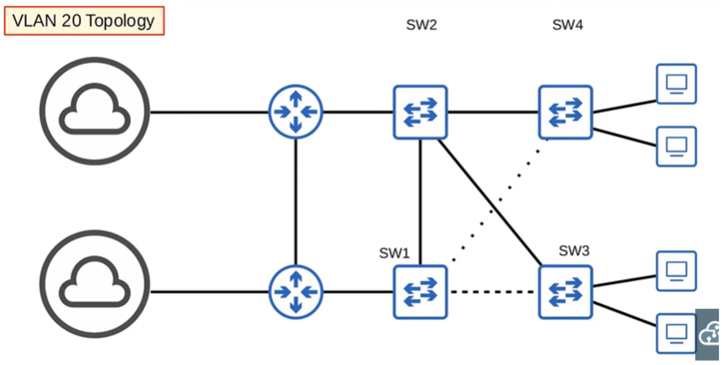
- SW2 is the root bridge for VLAN 20
- SW2 is backup root for VLAN 10

Resulting VLAN 10 Topology



- Traffic prefers paths toward **SW1**
- Some links are blocked to prevent loops

Resulting VLAN 20 Topology



- Traffic prefers paths toward **SW2**
- Different links are blocked compared to VLAN 10

Load Balancing Result

- VLAN 10 and VLAN 20 use **different interfaces**
- Traffic is distributed
- Interface bandwidth is not wasted

Spanning Tree Port Settings

```
SW2(config-if)#spanning-tree vlan 1 ?
  cost           Change an interface's per VLAN spanning tree path cost
  port-priority  Change an interface's spanning tree port priority

SW2(config-if)#spanning-tree vlan 1
```

- There are two main per-port settings:
  - Cost
  - Port Priority
- Both are configured per VLAN

Spanning Tree Cost

- Cost is the root path cost
- Examples:
  - FastEthernet → 19
  - GigabitEthernet → 4
- Used to:
  - Select the root port
  - Break ties for designated ports

Spanning Tree Port Priority

- Port priority is the first half of the Port ID
- It is the final tiebreaker in root port selection
- Used to influence port roles

Why Change Cost or Priority

- To change:
  - Root port selection
  - Designated / non-designated port selection

Interface Cost Configuration (CLI)

spanning-tree vlan 1 cost 200

- Cost range: **1 to 200,000,000**
- Applied per VLAN on the interface

```
SW2(config-if)#spanning-tree vlan 1 cost ?
  <1-200000000>  Change an interface's per VLAN spanning tree path cost

SW2(config-if)#spanning-tree vlan 1 cost 200
```

Interface Port Priority Configuration (CLI)

spanning-tree vlan 1 port-priority 32

- Priority range: **0 to 224**
- In increments of **32**
- Applied per VLAN

```
SW2(config-if)#spanning-tree vlan 1 port-priority ?
  <0-224>  port priority in increments of 32

SW2(config-if)#spanning-tree vlan 1 port-priority 32
```